Luis Lopez Rondon

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Education

University of South Florida — **Tampa, FL**
Bachelor of Science in Electrical Engineering — ***GPA: 4.00*** | *May 2029*

Hillsborough College — **Tampa, FL**
Associate of Arts — ***GPA: 4.00*** | *May 2026*

Relevant Coursework: Differential Equations, Digital Circuits

Honors: National Merit Finalist, Sunshine State Scholar, Judy Genshaft Honors College

Technical Skills

Hardware: PCB design, buck/boost converter PCBs, soldering (SMD, THT), Li-ion battery pack design
Embedded: ESP32, Teensy | I2C, SPI, CAN
EDA & Simulation: Altium Designer, KiCad, LTspice, Keysight ADS
Programming: C/C++, MATLAB
Spoken Languages: English, Spanish

Engineering Experience

IEEE at USF EXO — **Tampa, FL**
Electrical Systems Lead, Lower-Limb Exoskeleton — *Aug 2026 – Present*

- Developing the electrical system for a lower-limb exoskeleton driven by **48V CubeMars AK80-9 actuators**, including IMU and FSR sensing and DC-DC rails for the Jetson Orin Nano and Teensy 4.1.
- Created a schematic for a **hardware e-stop** using a relay to cut the motor bus while keeping control electronics powered, alongside a software e-stop for development testing.
- Developed a **MATLAB battery current model** to size the 48V pack (∼**10A peak, 3–4A mean** vs. >**50A** initial estimate) and performed a weighted benefit analysis of pre-built and custom Li-ion packs.

Projects

PID-Stabilized Parkinson's Spoon | *IEEE at USF EXO* — *Aug 2026 – Present*

- Designed the PCB for a PID-controlled Parkinson's spoon with a 1S LiPo, TP4056 charging, DW01A/FS8205 battery protection, and a custom **5V boost converter** for the servo power rail.
- Taught the design in an Altium workshop for ∼**10** EXO members, through their first PCB layout.

Mime: Mimetic Robotic Hand | *IEEE at USF EXO* — *May 2026 – Aug 2026*

- Designed a PCB (**ESP32-S3, PCA9685 I2C servo driver**) using **Altium Designer** for a robotic hand driven by a flex-sensor glove and **OpenCV**.
- Drove **5** MG996R servos from a dedicated **6V** supply, kept separate from the AMS1117 logic rail.

Audio Spectrum Analyzer | *Teensy 4.0, C++, PJRC Audio Library* — *Aug 2026 – Present*

- Built a real-time analyzer with a **1024-point** FFT in **16 log-spaced bands**, USB and line-in audio, and an ILI9341 TFT over SPI.
- Designing a custom PCB revision with discrete ADC/DAC circuits to add speaker output.

Work Experience

Tutoring Club of Westchase — **Tampa, FL**
Math and Reading Tutor — *Jan 2026 – Aug 2026*

- Tutored up to 3 K–12 students at a time in reading and math, from elementary material through AP Calculus AB, SAT, and ACT, focusing on reviewing mistakes and building problem-solving strategies.